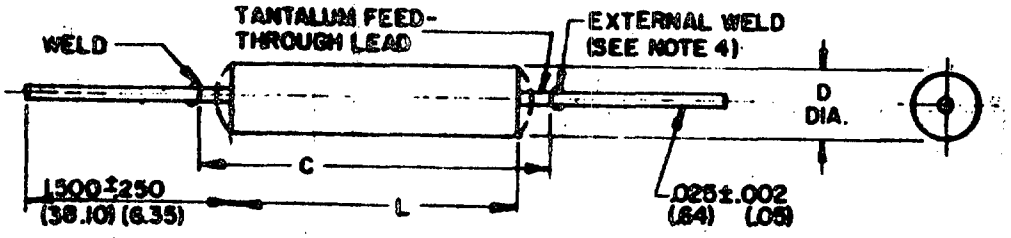
MILITARY SPECIFICATION SHEET

CAPACITORS, FIXED, ELECTROLYTIC (NONSOLID ELECTROLYTE), TANTALUM, (NONPOLARIZED, PLAIN FOIL) -85°C (VOLTAGE DERATED TO 125°C), STYLES CL32 AND CL33

INACTIVE FOR NEW DESIGN; USE MIL-C-39006/4

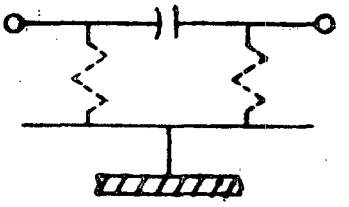
The complete requirements for procuring the capacitors described herein shall consist of this document and the latest issue of Specification MIL-C-3965.

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.



Case Size	Dimensions			
	L 1/ ±.062 (1.57)	D ±.016 (.41)	D Max	C Max
		CL32	CL33	
E1	.688 (17.48)	.188 (4.78)	.219 (5.56)	1.188 (30.18)
G2	.969 (24.61)	.281 (7.14)	.312 (7.92)	1.469 (37.31)
G3	1.438 (36.53)	.375 (9.53)	.406 (10.31)	1.938 (49.23)
G4	2.125 (53.98)	.375 (9.53)	.406 (10.31)	2.625 (66.68)
G5	2.750 (69.85)	.375 (9.53)	.406 (10.31)	3.250 (82.55)

1/ For style CL33, nonshrinkable sleeving shall extend .016 inch minimum, .062 inch maximum, beyond each end of the case; shrinkable sleeving shall lap over the ends of the case.



CIRCUIT DIAGRAM

NOTES:
1. Dimensions are in inches.
2. Millimeters are in parentheses.
3. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4mm.
4. The weld shall not be enclosed in the end seal.

FIGURE 1. Styles CL32 and CL33 Capacitors.

FSC 5910

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TABLE I. Capacitor characteristics.

Type designation 1/	Voltage			Capacitance	DC leakage (max)		Impedance (max)	Case size
	Rated (85°C)	Derated (125°C)	Surge (85°C)		25°C	85°C & 125°C		
	volts, dc	volts, dc	volts, dc	µF	µA	µA	ohms	
CL3-BA100MN E	3	2	3.4	10	1	2	283	E1
CL3-BA450MN G	3	2	3.4	45	2	4	59	G2
CL3-BA141MN G	3	2	3.4	140	2	10	19	G3
CL3-BA281MN G	3	2	3.4	280	3	20	10	G4
CL3-BA401MN G	3	2	3.4	400	6	30	6	G5
CL3-BB070MN E	6	4	6.9	7	1	2	377	E1
CL3-BB250MN G	6	4	6.9	25	2	4	105	G2
CL3-BB850MN G	6	4	6.9	85	2	10	31	G3
CL3-BB171MN G	6	4	6.9	170	4	20	15	G4
CL3-BB251MN G	6	4	6.9	250	5	30	11	G5
CL3-BD040MN E	10	7	11.5	4	1	2	660	E1
CL3-BD160MN G	10	7	11.5	16	2	4	165	G2
CL3-BD550MN G	10	7	11.5	55	2	10	48	G3
CL3-BD111MN G	10	7	11.5	110	4	20	24	G4
CL3-BD151MN G	10	7	11.5	150	5	30	18	G5
CL3-BE2R5MN E	15	10	17.2	2.5	1	2	893	E1
CL3-BE100MN G	15	10	17.2	10	2	4	223	G2
CL3-BE350MN G	15	10	17.2	35	2	10	64	G3
CL3-BE700MN G	15	10	17.2	70	4	20	32	G4
CL3-BE101MN G	15	10	17.2	100	5	30	23	G5
CL3-BG1R5MN E	25	15	28.8	1.5	1	3	1487	E1
CL3-BG060MN G	25	15	28.8	6	2	6	372	G2
CL3-BG200MN G	25	15	28.8	20	2	10	112	G3
CL3-BG400MN G	25	15	28.8	40	4	20	56	G4
CL3-BG600MN G	25	15	28.8	60	5	30	37	G5
CL3-BH1R4MN E	30	20	34.5	1.4	2	5	1594	E1
CL3-BH5R5MN G	30	20	34.5	5.5	2	6	496	G2
CL3-BH180MN G	30	20	34.5	18	2	12	124	G3
CL3-BH360MN G	30	20	34.5	36	4	26	62	G4
CL3-BH450MN G	30	20	34.5	45	6	36	50	G5
CL3-BJ0R8MN E	50	30	57.5	0.8	1	4	2790	E1
CL3-BJ030MN G	50	30	57.5	3	2	6	743	G2
CL3-BJ100MN G	50	30	57.5	10	2	12	223	G3
CL3-BJ200MN G	50	30	57.5	20	4	24	112	G4
CL3-BJ300MN G	50	30	57.5	30	6	36	75	G5
CL3-BL0R5MN E	75	50	86.2	0.5	1	4	4460	E1
CL3-BL020MN G	75	50	86.2	2	2	6	1115	G2
CL3-BL070MN G	75	50	86.2	7	2	13	319	G3
CL3-BL140MN G	75	50	86.2	14	4	24	159	G4
CL3-BL200MN G	75	50	86.2	20	6	36	112	G5
CL3-BN0R4MN E	100	65	115.0	0.4	1	3.5	5580	E1
CL3-BN1R5MN G	100	65	115.0	1.5	2	6	1487	G2
CL3-BN050MN G	100	65	115.0	5	2	13	446	G3
CL3-BN100MN G	100	65	115.0	10	4	24	223	G4
CL3-BN150MN G	100	65	115.0	15	6	36	149	G5
CL3-BQR25MN E	150	100	172.0	0.25	2	4	8908	E1
CL3-BQ010MN G	150	100	172.0	1	2	6	2227	G2
CL3-BQ3R5MN G	150	100	172.0	3.5	3	13	638	G3
CL3-BQ070MN G	150	100	172.0	7	4	24	319	G4
CL3-BQ100MN G	150	100	172.0	10	6	36	223	G5

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TABLE I. Capacitor characteristics. -Continued

Type designation 1/	Voltage			Capacitance	DC leakage (max)		Impedance (max)	Case size
	Rated (85°C)	Derated (125°C)	Surge (85°C)		25°C	85°C & 125°C		
	volts, dc	volts, dc	volts, dc	µF	µA	µA	ohms	
CL3-BZR15LN E	200	150	230.0	0.15	4	25	14000	E1
CL3-BZR75LN G	200	150	230.0	0.75	8	32	2600	G2
CL3-BZ2R5LN G	200	150	230.0	2.5	16	64	800	G3
CL3-BZ050LN G	200	150	230.0	5	28	112	400	G4
CL3-BZ7R5LN G	200	150	230.0	7.5	40	160	260	G5
CL3-BOR15LN E	250	165	287.0	0.15	6	32	13000	E1
CL3-BOOR6LN G	250	165	287.0	0.6	8	40	3300	G2
CL3-BO020LN G	250	165	287.0	2	16	80	1000	G3
CL3-BO040LN G	250	165	287.0	4	28	140	500	G4
CL3-BO060LN G	250	165	287.0	6	40	200	330	G5
CL3-BXR12LN E	300	200	345.0	0.12	4	25	17500	E1
CL3-BXR47LN G	300	200	345.0	0.47	10	50	4500	G2
CL3-BX1R5LN G	300	200	345.0	1.5	20	100	1500	G3
CL3-BX3R3LN G	300	200	345.0	3.3	35	175	643	G4
CL3-BX4R7LN G	300	200	345.0	4.7	50	250	450	G5
CL3-BY0R1LN E	375	250	431.0	0.1	8	80	23500	E1
CL3-BYR39LN G	375	250	431.0	0.39	10	100	5620	G2
CL3-BY1R2LN G	375	250	431.0	1.2	20	225	1880	G3
CL3-BY2R2LN G	375	250	431.0	2.2	35	250	900	G4
CL3-BY3R9LN G	375	250	431.0	3.9	50	325	562	G5

1/ Complete type designation will include an additional symbol to indicate style.

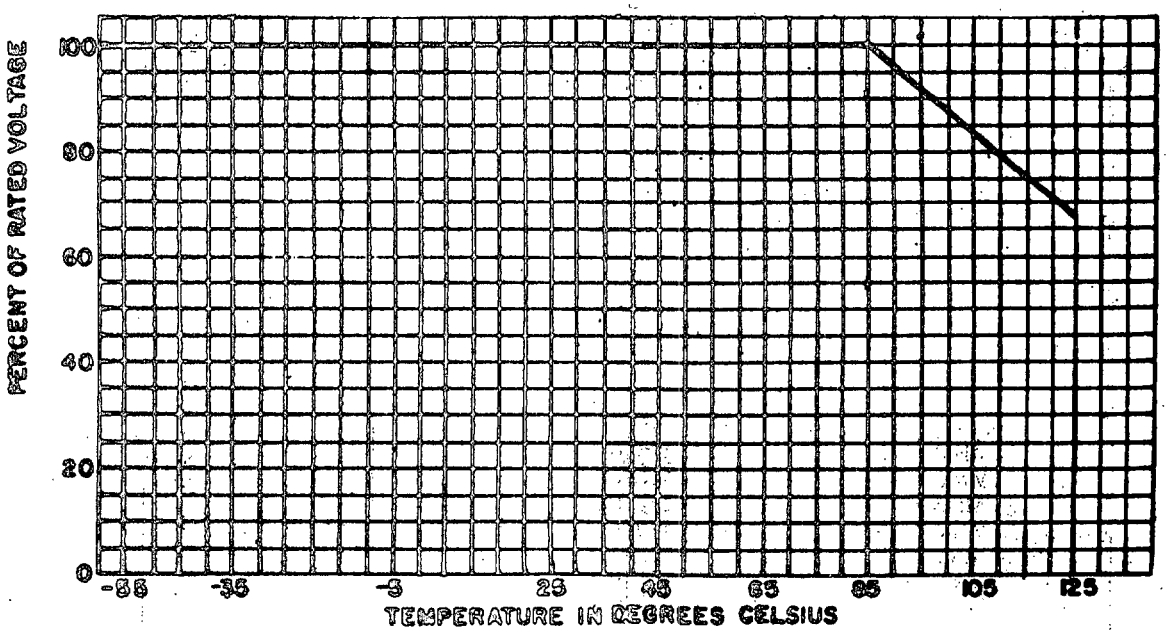


FIGURE 2. Voltage derating with temperature.

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TABLE II SUPERSESSION & CROSS REFERENCE DATA

SUPERSEDED		
FROM Type Designation[1/] MIL-C-3965/16C	TO Type Designation[1/] MIL-C-3965/16D	Cross Reference For New Design Part Number MIL-C-39006/4C
CL3-BA100MNE	CL3-BA100MNE	No Replacement
CL3-BA450MNE	CL3-BA450MNE	No Replacement
CL3-BA141MNE	CL3-BA141MNG	No Replacement
CL3-BA281MNE	CL3-BA281MNG	No Replacement
CL3-BA401MNE	CL3-BA401MNG	No Replacement
CL3-BB070MNE	CL3-BB070MNE	No Replacement
CL3-BB250MNE	CL3-BB070MNG	No Replacement
CL3-BB850MNE	CL3-BB850MNG	No Replacement
CL3-BB171MNE	CL3-BB171MNG	No Replacement
CL3-BB251MNE	CL3-BB251MNG	No Replacement
CL3-BD040MNE	CL3-BD040MNE	No Replacement
CL3-BD160MNE	CL3-BD160MNG	No Replacement
CL3-BD550MNE	CL3-BD550MNG	No Replacement
CL3-BD111MNE	CL3-BD111MNG	No Replacement
CL3-BD151MNE	CL3-BD151MNG	No Replacement
CL3-BE2R5MNE	CL3-BE2R5MNE	M39006/04-2355
CL3-BE100MNE	CL3-BE100MNG	M39006/04-1245
CL3-BE350MNE	CL3-BE350MNG	M39006/04-1246
CL3-BE700MNE	CL3-BE700MNG	M39006/04-1247
CL3-BE101MNE	CL3-BE101MNG	M39006/04-1248
CL3-BG1R5MNE	CL3-BG1R5MNE	M39006/04-2360
CL3-BG060MNE	CL3-BG060MNG	M39006/04-1249
CL3-BG200MNE	CL3-BG200MNG	M39006/04-1250
CL3-BG400MNE	CL3-BG400MNG	M39006/04-1251
CL3-BG600MNE	CL3-BG600MNG	M39006/04-1252
CL3-BH1R4MNE	CL3-BH1R4MNE	M39006/04-2365
CL3-BH5R5MNE	CL3-BH5R5MNG	M39006/04-1253
CL3-BH180MNE	CL3-BH180MNG	M39006/04-1254
CL3-BH360MNE	CL3-BH360MNG	M39006/04-1255
CL3-BH450MNE	CL3-BH450MNG	M39006/04-1256
CL3-BJ0R8MNE	CL3-BJ0R8MNE	M39006/04-2370
CL3-BJ030MNE	CL3-BJ030MNG	M39006/04-1257
CL3-BJ100MNE	CL3-BJ100MNG	M39006/04-1258
CL3-BJ200MNE	CL3-BJ200MNG	M39006/04-1259
CL3-BJ300MNE	CL3-BJ300MNG	M39006/04-1260
CL3-BL0R5MNE	CL3-BL0R5MNE	M39006/04-2375
CL3-BL020MNE	CL3-BL020MNG	M39006/04-1261
CL3-BL070MNE	CL3-BL070MNG	M39006/04-1262
CL3-BL140MNE	CL3-BL140MNG	M39006/04-1263
CL3-BL200MNE	CL3-BL200MNG	M39006/04-1264
CL3-BN0R4MNE	CL3-BN0R4MNE	M39006/04-2380
CL3-BN1R5MNE	CL3-BN1R5MNG	M39006/04-1265
CL3-BN050MNE	CL3-BN050MNG	M39006/04-1266
CL3-BN100MNE	CL3-BN100MNG	M39006/04-1267
CL3-BN150MNE	CL3-BN150MNG	M39006/04-1268
CL3-BQR25MNE	CL3-BQR25MNE	M39006/04-2385
CL3-BQ010MNE	CL3-BQ010MNG	M39006/04-1269
CL3-BQ3R5MNE	CL3-BQ3R5MNG	M39006/04-1271
CL3-BQ070MNE	CL3-BQ070MNG	M39006/04-1272
CL3-BQ100MNE	CL3-BQ100MNG	M39006/04-1273
CL3-BZR15LNE	CL3-BZR15LNE	M39006/04-2390
CL3-BZR75LNE	CL3-BZR75LNG	M39006/04-1274
CL3-BZ2R5LNE	CL3-BZ2R5LNG	M39006/04-1275
CL3-BZ050LNE	CL3-BZ050LNG	M39006/04-1276
CL3-BZ7R5LNE	CL3-BZ7R5LNG	M39006/04-1277
CL3-BOR15LNE	CL3-BOR15LNE	M39006/04-2395
CL3-B00R6LNE	CL3-B00R6LNG	M39006/04-1278
CL3-B0020LNE	CL3-B0020LNG	M39006/04-1279
CL3-B0040LNE	CL3-B0040LNG	M39006/04-1280
CL3-B0060LNE	CL3-B0060LNG	M39006/04-1281

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TABLE II SUPERSESSION & CROSS REFERENCE DATA (Cont'd)

SUPERSEDED		
FROM Type Designation[1]/ MIL-C-3965/16C	TO Type Designation[1]/ MIL-C-3965/16D	Cross Reference For New Design Part Number MIL-C-39006/4C
CL3-BXR47LNE	CL3-BXR47LNG	M39006/04-1282
CL3-BX1R5LNE	CL3-BX1R5LNG	M39006/04-1283
CL3-BX3R3LNE	CL3-BX3R3LNG	M39006/04-1284
CL3-BX4R7LNE	CL3-BX4R7LNG	M39006/04-1285
CL3-BYOR1LNE	CL3-BYOR1LNE	M39006/04-2405
CL3-BYR39LNE	CL3-BYR39LNG	M39006/04-1286
CL3-BY1R2LNE	CL3-BY1R2LNG	M39006/04-1287
CL3-BY2R9LNE	CL3-BY2R9LNG	M39006/04-1288
CL3-BY3R9LNE	CL3-BY3R9LNG	M39006/04-1289

1/ Complete Type Designation will include an additional symbol to indicate style.

REQUIREMENTS:

Design and construction:
Dimensions and configuration - See Fig 1.
Case type - Tubular; uninsulated (CL32), insulated (CL33).
Seal type - Hermetic (symbol G).
- Nonhermetic (Symbol E).
Terminals - Axial wire-lead.
Rated temperature - -55°C to +85°C, voltage derated to +125°C (See Fig 2).
Rated voltage - See Table I.
Capacitance (cap) (nom) - See Table I.
Capacitance tolerance - Symbol L (±15%) or M (±20%) as shown in Table I.

DC leakage (DCL) (max) (at +25°C):
See Table I.

Dissipation factor (DF) (max):
15% (≤10v) or 10% (≥15v).

Seal: In accordance with MIL-C-3965.

Dielectric withstanding voltage (DWV):
Sleeving to case (min) - 2,000 vdc.
Terminal to case - Not applicable.

Insulation resistance:
Sleeving to case (min): 100 megohms.
Terminal to case - Not applicable.

Temperature cycling:
DCL (max) - 125% of 25°C value of Table I.
Δ Cap - Within ±5% of initial measurement.
DF (max) - 115% of initial requirement.

Solderability: In accordance with MIL-C-3965.

Terminal strength:
Pull test-In accordance with MIL-C-3965.
Bend test - Not applicable.
Torque test - Not applicable.
Wire-lead bend test - In accordance with MIL-C-3965.
Twist test - Not applicable.

1/ 125°C derated voltage referenced.

Surge voltage: See Table I.

Moisture resistance:
DCL(max) - 125% of 25°C value of Table I.
Δ Cap - Within ±3% of initial measurement.
DF (max) - 115% of initial requirement.

Low temperature (storage):
DCL (max) - See Table I (25°C).
Δ Cap - Within ±3% of initial measurement.
DF (max) - Shall not exceed initial requirement.

Stability at low and high temperature:
Step 1 (+25°C):
DCL (max) - See Table I.
Cap - Within tolerance of Table I value.
DF (max) - Shall not exceed initial requirement.
Step 2 (-55°C):
Impedance - See Table I.
Δ Cap (max) - -25% (≤250v) or -40% (>250v) of step 1 value.
Step 3 (+25°C):
DCL (max) - See Table I.
Δ Cap - Within +3% of step 1 value.
DF (max) - Shall not exceed initial requirement.
Step 4 (+85°C):
DCL (max) - See Table I.
Δ Cap (max) - +20% (≤50v) or +15% (>50v) of step 1 value.
DF (max) - 20% (≤50v) or 15% (>50v).
1/ Step 5 (+125°C):
DCL (max) - See Table I.
Δ Cap (max) - +25% (≤30v) or +15% (>30v) of step 1 value.
DF (max) - 25%(≤30v) or 15%(>30v).
Step 6 (+25°C):
DCL (max) - See Table I.
Δ Cap-Within ±3% of initial measurement.
DF(max)-Shall not exceed initial requirement.

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Reverse voltage: Not applicable.

Life at +85°C:
2,000-hr (qualification).
DCL (max) - See Table I (25°C).
ΔCap - Within ±15% (≤50v) or ±10% (>50v) of initial measurement.
DF (max) - Shall not exceed initial requirement.
240-hr (Group B):
DCL (max) - See Table I (25°C).
ΔCap - Within ±10% of initial measurement.
DF (max) - Shall not exceed initial requirement.
1,760-hr (Group C):
DCL (max) - See Table I (25°C).
ΔCap - Within ±15% (≤50v) or ±10% (>50v) of initial measurement.
DF (max) - Shall not exceed initial requirement.

Barometric pressure:
Test voltage - Rated voltage.

1/ Life at +125°C:
DCL (max) - See Table I (25°C).
ΔCap - Within ±15% (≤30v) or ±10% (>30v) of initial measurement.
DF (max) - Shall not exceed initial requirement.
DWV (min)(CL33 only) - 2,000 vdc.
IR (min) (CL33 only) - 100 megohms.

Voltage groups (85°C rated voltage):
I---------- 15 to 50v, incl.
II---------- 75 to 200 v, incl.
III--------- 250 to 375v.

1/ 125°C derated voltage referenced.

NOTE: The capacitors covered by this specification sheet are substitutable for those covered by MIL-C-3965/16C. See Table II.

Custodians:
Army - EL
Navy - EC
Air Force 80

Review activities:
Army - EL, MU
Navy - EC, SH
Air Force - 11, 17
DSA - ES

User Activities:
Army -
Navy - AS, MC, OS
Air Force - 19

Preparing Activity:
Air Force - 80

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